

F5 Distributed Cloud DNS - Lab Guide Series



Overview

This repository contains comprehensive hands-on lab guides for **F5 Distributed Cloud DNS Training**. Learn how to leverage F5 XC DNS for high availability, DDoS protection, and secure DNS architecture.

Item	Details
Course	F5 Distributed Cloud DNS Training
Lab Environment	Kasm Workspaces (Container Streaming Platform)
UDF Environment	xc-dns-lab-v1
Requirements	NO LIVE TENANT REQUIRED

Getting Started

Before starting the labs, you need to access the lab environment through F5 UDF and Kasm Workspaces.

 [How to Access Lab Environment](#) — Step-by-step guide to access UDF and Kasm Workspaces

Lab Series

Lab	Topic	Description	Duration
Lab 1	DNS Availability	Eliminate single-provider DNS dependency using F5 XC DNS as a dual primary provider	~25 min
Lab 2	Attack Mitigation	XC DNS absorbs DDoS attacks — watch a live DNS flood get mitigated in real time	~30 min
Lab 3	Hidden Primary	Hide BIND behind XC DNS for maximum protection — attackers can't target what they can't find	~30 min

Lab Details

Lab 1 — DNS Availability

Objective: Eliminate single-provider DNS dependency using F5 Distributed Cloud DNS as a dual primary provider.

What You'll Learn:

- ✔ Verify baseline DNS configuration with Legacy BIND

-  Simulate DNS provider outage and observe impact
-  Create DNS zone in F5 XC Console
-  Configure NS delegation at Registrar
-  Implement dual primary DNS architecture

 [View Lab 1 Guide](#) |  [PDF Version](#)

Lab 2 — Attack Mitigation

Objective: Demonstrate how XC DNS absorbs DDoS attacks while Legacy BIND would be overwhelmed.

What You'll Learn:

-  Monitor healthy DNS baseline
-  Launch simulated DNS flood attack
-  Observe attack impact on infrastructure
-  See XC DNS absorb and mitigate the attack
-  Verify service continuity during attack

 [View Lab 2 Guide](#) |  [PDF Version](#)

Lab 3 — Hidden Primary

Objective: Implement hidden primary architecture where BIND is completely invisible to attackers.

What You'll Learn:

-  Understand hidden primary DNS architecture
-  Configure zone transfer (AXFR) from BIND to XC DNS
-  Remove BIND NS records from public delegation
-  Verify automatic zone synchronization
-  Test attack resilience with hidden primary

 [View Lab 3 Guide](#) |  [PDF Version](#)

Repository Structure

```
f5-xc-dns-lab-guide/
├── README.md                # This file
├── how to access lab environment/ # Getting Started Guide
│   ├── How_to_Access_Lab_Environment.md
│   └── step*.png            # Access guide screenshots
├── lab1/                    # Lab 1: DNS Availability
│   ├── F5_XC_DNS_Lab_Guide.md
│   ├── F5_XC_DNS_Lab_Guide.pdf
│   └── step*.png            # Step-by-step screenshots
└── lab2/                    # Lab 2: Attack Mitigation
```

```
├── F5_XC_DNS_Lab2_Attack_Mitigation_Guide.md
├── F5_XC_DNS_Lab2_Attack_Mitigation_Guide.pdf
├── step*.png          # Step-by-step screenshots
└── lab3/              # Lab 3: Hidden Primary
    ├── F5_XC_DNS_Lab3_Hidden_Primary_Guide.md
    ├── F5_XC_DNS_Lab3_Hidden_Primary_Guide.pdf
    └── step*.png      # Step-by-step screenshots
```

💡 Why No Live Tenant Required?

This lab uses a **fully simulated F5 XC DNS environment** that provides the same hands-on experience as a live tenant:

Feature	Benefit
🖥️ Realistic XC Console Interface	Learn the real workflow and navigation
🌐 Simulated DNS Infrastructure	All operations behave exactly like production
🔒 Safe Learning Environment	Make mistakes without affecting real infrastructure
⚡ No Account Setup Required	Jump straight into learning
🎯 Consistent Experience	Every participant gets the same pre-configured environment

🔒 Lab Progression

Labs are designed to be completed **sequentially**. Each lab builds upon the configuration and knowledge from the previous lab:

```
Lab 1 (DNS Availability)
|
▼ Requires dual primary setup
Lab 2 (Attack Mitigation)
|
▼ Requires understanding of availability + mitigation
Lab 3 (Hidden Primary)
```

💡 **Tip:** Complete Lab 1 first, then Lab 2 and Lab 3 will automatically unlock.

🔧 Prerequisites

- Modern web browser (Chrome recommended)
- Access to Kasm Workspaces environment
- Credentials provided by your instructor

📞 Support

If you encounter any issues during the lab, please contact your instructor.



License

This training material is provided for educational purposes as part of F5 Distributed Cloud DNS Training.

Created for F5 Distributed Cloud DNS Training

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